

Acknowledgements.

Contents

Introduction	1
1 Theoretical Framework	2
1.1 The standard Black–Scholes framework	2
1.2 Why constant volatility is restrictive	3
1.3 Main directions beyond the Black–Scholes benchmark	3
1.4 Regime-switching models	4
1.5 Empirical state identification under the physical measure	5
2 Data and Empirical Methodology	6
2.1 Data construction and validation	6
2.2 Exploratory analysis	6
2.3 Gaussian Hidden Markov specification	8
2.4 Estimation strategy and parameter counting	8
2.5 Model comparison criteria and regime labelling	9
3 Empirical Results	10
3.1 Comparison of the two-state and three-state models	10
3.2 The two-state classification	11
3.3 The three-state classification	11
3.4 Transition matrices and regime persistence	13
3.5 Continuous-time approximation of switching intensities	13
4 Option Pricing Application	15
4.1 Pricing exercise and walk-forward design	15
4.2 Walk-forward regime inputs	16
4.3 Regime-switching Black–Scholes pricing equation	17
4.4 Numerical solution	18
4.5 Option data and pricing assumptions	18
4.6 Pricing results	19
4.7 Comparison with a Black–Scholes benchmark	21
4.8 Implied-volatility comparison	23
4.9 Interpretation and limitations	24
5 Out-of-the-money option pricing	25
5.1 OTM implied-volatility comparison	26
5.2 Walk-forward OTM volatility adjustment	27
5.3 Overall interpretation of the OTM exercise	29
6 Conclusion	30

Introduction

The Black–Scholes model is crucial in modern financial theory due to the fact that it provides a closed-form solution of the price of European option in a transparent no-arbitrage framework. As a result, it can be considered as a benchmark in the development of other theoretical models of option pricing. Given its importance, it is vital to discuss both its strengths and limitations arising from its assumptions, which contradict empirical findings.

A major restriction of the model is fixed volatility. Time-varying volatility is a common phenomenon for stock index markets, particularly for the S&P 500. Quiet periods alternate with periods of instability. Large changes in price tend to occur in clusters, not in isolation. Options are sensitive to volatility changes, so assuming a constant volatility parameter could make mathematical calculations more straightforward, but may not reflect volatility changes in the real world. Such models are simpler to work with technically but may struggle in modeling the movement of markets.

A natural extension is therefore to allow volatility to vary with market conditions. In regime-switching models, volatility depends on latent states that are not directly observed but are inferred from the data. These frameworks encompass just a small number of different modes, but represent different patterns in market behavior. Each state is associated with its own level of volatility and with transition probabilities governing movements across regimes. The main advantage of this approach is its ease of depicting complex patterns. Transitions generate periods of calm, moderate fluctuation, or turbulence. The structure remains relatively simple while offering an understanding of the evolution of markets over time.

This thesis models such regimes using Hidden Markov Models, which are not directly observed, but inferred from daily returns of the S&P 500. In the empirical analysis, the two- and three-state Gaussian HMMs are compared in terms of their fit, the regimes are ranked in terms of volatility, and the persistence of the HMMs is examined using the inferred transition matrix and continuous-time generator approximation.

The thesis is organized as follows. In the first chapter, the theoretical framework is introduced, starting with the standard Black–Scholes model and its key assumptions, and then the discussion focuses on the restrictive nature of the constant-volatility hypothesis in practice. It also introduces regime-switching models as a natural extension in cases of regime switching market volatility. The dataset and empirical methods are described in Chapter 2 with regard to data construction, exploratory analysis and specification of the Gaussian Hidden Markov Models. The empirical results are presented in chapter 3, where the two-state and three-state specifications are compared as well as regime persistence investigated by estimated transition matrices and continuous-time generator approximations. Chapter 4 applies the walk-forward regime-switching Black–Scholes model to at-the-money options to compare with a rolling-volatility Black–Scholes benchmark. Chapter 5 extends the analysis to OTM options and examines the potential benefits of an IV-based adjustment on pricing.

Chapter 1

Theoretical Framework

1.1 The standard Black–Scholes framework

Let S_t denote the asset price at time t . Under the physical measure $\mathbb{P}$, the model assumes that the price follows a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad (1.1)$$

where μ is the expected rate of return, $\sigma > 0$ is a constant volatility parameter, and $W_t^{\mathbb{P}}$ is a standard Brownian motion. As a result, log returns follow a normal distribution.

A key feature of the model is that a continuously rebalanced hedging portfolio can be constructed to eliminate local risk. Under the absence of arbitrage, this leads to a risk-neutral representation in which the drift of the asset is replaced by the risk-free rate r . Under the pricing measure $\mathbb{Q}$, the asset dynamics become

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (1.2)$$

For a derivative price $V(S, t)$, the no-arbitrage argument yields the Black–Scholes partial differential equation

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0. \quad (1.3)$$

The assumptions behind this derivation are strong and should be stated clearly. In its simplest form, the model assumes frictionless markets, no arbitrage, continuous trading, unlimited short selling, a constant risk-free interest rate, and a non-dividend-paying underlying asset. It also assumes that volatility is constant and known. For the purposes of the present thesis, the standard Black–Scholes call and put formulas without dividend adjustments are presented as the benchmark pricing framework.

1.1.1 European call and put pricing formulas

For a European call option with strike K and maturity T , the Black–Scholes price at time 0 is

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2), \quad (1.4)$$

while the price of a European put is

$$P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1), \quad (1.5)$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function and

$$d_1 = \frac{\log(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (1.6)$$

These formulas remain central in finance for two reasons. First, they deliver immediate and interpretable comparative statics: option values increase with volatility, maturity, and, in the case of calls, the underlying price. Second, they provide a benchmark against which model extensions can be judged. Even when a more sophisticated specification is ultimately preferred, Black–Scholes remains the natural starting point for both theoretical discussion and empirical implementation.

1.2 Why constant volatility is restrictive

Despite its importance, the Black–Scholes model is limited by its assumption of constant volatility. In practice, financial markets exhibit time-varying volatility, which may change gradually or shift abruptly during periods of stress. Calm periods are often followed by episodes of turbulence, and elevated volatility tends to persist rather than disappearing immediately. These features are clearly visible in stock-index return data and are difficult to reconcile with a constant-volatility specification.

A central issue here is how volatility shapes option pricing fundamentally. Far from being just another factor, it directly influences option values. When a model applies constant volatility to returns that actually shift between calm and turbulent phases, mismatches emerge, distorting both asset behavior and price patterns. Another issue arises when comparing such models to real-world data: observed implied volatilities change based on strike price and time to expiration. This variation contradicts the idea that one fixed volatility fits all scenarios. So instead of converging into uniform values, market signals show dispersion across different contracts.

For the present thesis, the problem is even more direct. The empirical part studies daily S&P 500 returns over a ten-year period. When markets are calm, their fluctuations behave differently compared to moments of intense pressure, this difference matters. Because of such shifts, using a model where volatility changes with market conditions is not merely a technical refinement. Instead, it aligns the pricing mechanism more closely with what actual patterns in the data show.

1.3 Main directions beyond the Black–Scholes benchmark

The literature has proposed several routes beyond constant volatility. Local-volatility models allow the diffusion coefficient to depend on the price level and time. Stochastic-volatility models treat volatility itself as a latent continuous-time process. Jump-diffusion models enrich the return dynamics by admitting discontinuous price movements. Each of these approaches relaxes the benchmark in a different way, and each has analytical and computational costs. Regime-switching models occupy an intermediate position. Rather than treating volatility as a continuously evolving latent process, these frameworks suppose markets jump between distinct phases. Each phase persists over time, bringing along its unique level of turbulence. One appeal lies in simplicity, a parsimonious structure. Another stems from how closely such shifts mirror real financial turning points. What stands out first is how it reflects changing volatility through an economically meaningful lens. Following that, a fixed number of states emerges, this aligns smoothly with Markov-type behavior and linked pricing models.

1.4 Regime-switching models

A regime-switching model introduces an unobserved state variable X_t taking values in a finite set $\{1, \dots, m\}$. The state follows a Markov chain, so the probability of moving to the next state depends only on the current one. In a financial context, the state can affect one or several coefficients of the asset-price dynamics, including the drift, the volatility, and potentially the interest rate. In the present thesis, volatility is the key coefficient of interest.

The economic interpretation is straightforward. A low-volatility regime may be associated with relatively stable financial conditions, abundant liquidity, and limited market stress. A medium-volatility regime may correspond to periods of uncertainty in which price fluctuations are larger but not yet extreme. A high-volatility regime may capture rare stress episodes, sharp repricing, or crisis-like conditions. The model does not require these labels to be observed in advance; they are inferred statistically from the data.

1.4.1 Economic meaning of volatility regimes

One reason traders find volatility regimes useful lies in how they simplify complex financial dynamics into clear states. Not every phase of trading looks the same, conditions shift over time. During some periods, confusion grows; at others, investors feel more confident, reacting to broader economic strains. Rather than treat each sudden change as random noise, models with limited states see patterns that last. These are not fleeting moments but phases that stick around for a while.

Most likely, this view fits well with wide stock market indexes. During long stretches of small price changes, markets seem quiet. Large price movements tend to cluster, suggesting another pattern takes hold. Such shifts become clearer when using a model that switches states, measuring how long each phase tends to last.

1.4.2 A regime-switching Black–Scholes setting

A conceptual regime-switching extension of Black–Scholes can be written by allowing volatility to depend on the latent state. Under the simplest risk-neutral specification,

$$dS_t = rS_t dt + \sigma_{X_t} S_t dW_t^Q, \quad (1.7)$$

where X_t is a finite-state Markov chain and σ_{X_t} takes one of the regime-specific values $\sigma_1, \dots, \sigma_m$. If the chain is modeled in continuous time, its dynamics are summarized by a generator matrix $Q = (q_{ij})$, where $q_{ij} \geq 0$ for $i \neq j$ and each row sums to zero.

If $V_i(S, t)$ denotes the option value when the current regime is i , the pricing problem becomes a system of coupled partial differential equations rather than a single equation. With a common risk-free rate r , the system takes the form

$$\frac{\partial V_i}{\partial t} + \frac{1}{2} \sigma_i^2 S^2 \frac{\partial^2 V_i}{\partial S^2} + rS \frac{\partial V_i}{\partial S} - rV_i + \sum_{j \neq i} q_{ij} (V_j - V_i) = 0, \quad i = 1, \dots, m. \quad (1.8)$$

The additional coupling term captures the fact that the option value in one state depends not only on the local Black–Scholes operator for that state, but also on the possibility of switching into other states. When $m = 1$, or when all regimes collapse to the same volatility, the system reduces to the standard Black–Scholes case.

1.5 Empirical state identification under the physical measure

Although the model is estimated from data, the Hidden Markov Model here does not deliver complete price alignment. Estimation relies on past returns, tying it to real-world $\mathbb{P}$ observations. Regime likelihoods inferred from historical returns reflect historical market dynamics, not how they must move under arbitrage-free rules $\mathbb{Q}$. What emerges describes history, not the adjusted probabilities needed for fair pricing.

This distinction is important. Instead of assuming uniform behavior, models reacting to shifts in conditions demand either clear rules under valuation settings or a defined bridge connecting real-world patterns to risk-adjusted ones, possibly via how markets value changes in regime likelihood. Looking back at past phases then makes most sense when seen ahead of time. Such analysis suggests how many distinct modes exist, captures varying swings tied to each phase, tracks how long states tend to last. Each piece fits directly into what follows: adaptation, sharpening, or new context within frameworks built around neutral expectations.

Chapter 2

Data and Empirical Methodology

2.1 Data construction and validation

Every entry has the following attributes recorded: the date, the value of S&P 500 Index, the U.S. Treasury rate for 3 months, the strike price of the option, the expiry of the option contract, and the call option price. These fields appear in the dataset under the labels `date`, `spx`, `r_3m`, `strike`, `expiry_date`, and `option_price`. Starting from price levels, daily changes are calculated using logarithms, then saved as `log_return`. Return series serves as the primary variable for the Hidden Markov Model, while option variables are considered when comparing model prices to observed prices.

Table 2.1: Variables used in the empirical analysis.

Variable	Description	Use in the thesis
<code>date</code>	Trading date	Time index
<code>spx</code>	S&P 500 index level	Underlying price series
<code>r_3m</code>	3-month U.S. Treasury bill rate	Risk-free rate proxy
<code>log_return</code>	$\log(S_t/S_{t-1})$	HMM estimation variable

Starting from 5 January 2015 up to 31 December 2024, the sample is made up of 2498 observations. With minimal preprocessing applied, dates are interpreted as time stamps; meanwhile, `spx` and `r_3m` undergo conversion into numerical form. Timestamps are sorted by date for chronological ordering, following parsing. It is also checked whether there are any gaps or duplicate timestamps in the sample during parsing. Data validation showed that no observations needed to be excluded as there were no duplicate values present. Calculating returns involves using previous values and, therefore, the first log difference is impossible. Daily log returns are defined as

$$R_t = \log \left(\frac{S_t}{S_{t-1}} \right). \quad (2.1)$$

Found often in real-world financial studies, this method fits well with the ongoing time model shown earlier in Chapter 1. Since the Hidden Markov Model considers daily data, the first step involves calculating the variation using daily figures to get annual results by multiplying by a factor of 252 trading days:

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_{\text{daily}} \sqrt{252}. \quad (2.2)$$

2.2 Exploratory analysis

Looking first at the data, visual inspection helps judge if constant volatility makes sense before applying Hidden Markov Models. The plot of daily log-returns reveals short-term fluctuations over time. Instead of symmetry, the distribution shows frequent small changes alongside rare large swings, visible in the return histogram. Another view emerges from the 21-day rolling measure, which tracks how annualized volatility shifts across periods.

Log returns on most days are more or less constant, although the variance tends to vary

with time. Large price movements tend to cluster rather than occur in isolation. Periods of low volatility may suddenly be followed by periods of market stress. Such occurrences raise attention regarding the factors that influence the variation in volatilities.

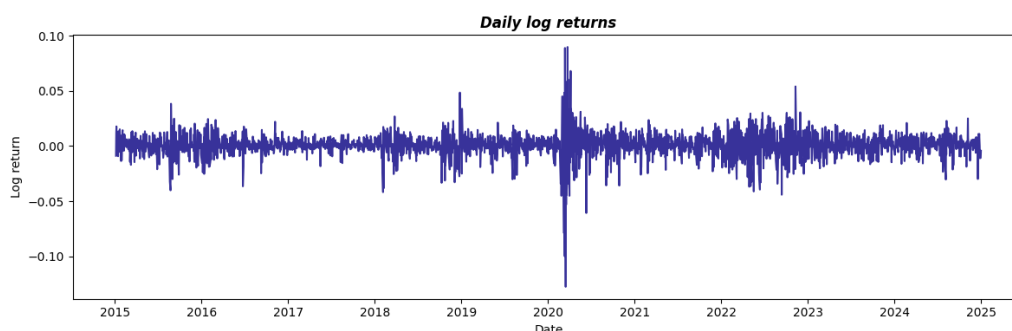


Figure 2.1: Daily log returns of the S&P 500 over the sample period.

The histogram confirms what the time-series presents. Around zero, the data points are concentrated, but other values go beyond that in both directions. In light of this data arrangement, using a single bell-shaped curve with fixed dispersion will fall short.

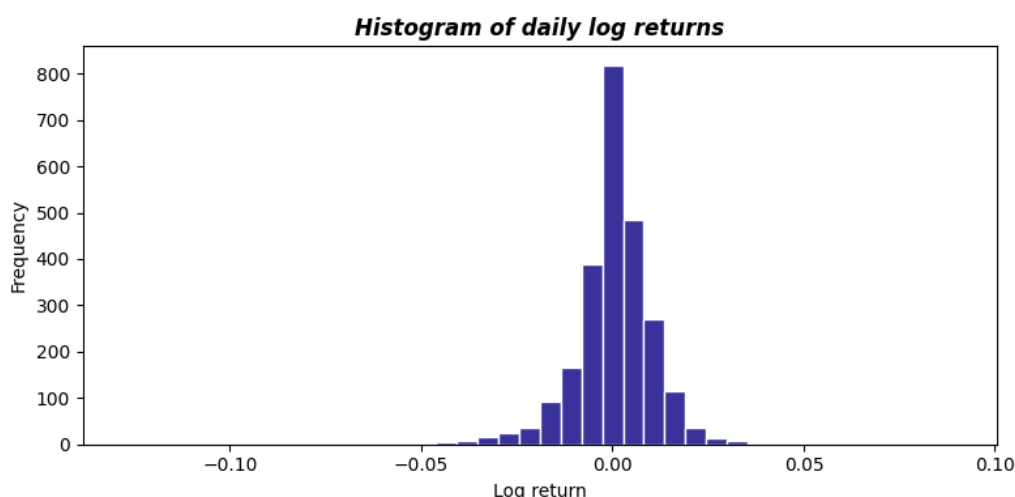


Figure 2.2: Histogram of daily S&P 500 log returns.

With time, it becomes increasingly clear where changes in risks become evident. While largely stable, it is clear that at certain instances, there is always an abrupt change that leads to disruption in the pattern of volatility. This supports the idea that volatility occurs in phases and is not just a reaction to a random shock. It is this element that is reflective of the key idea in this paper.

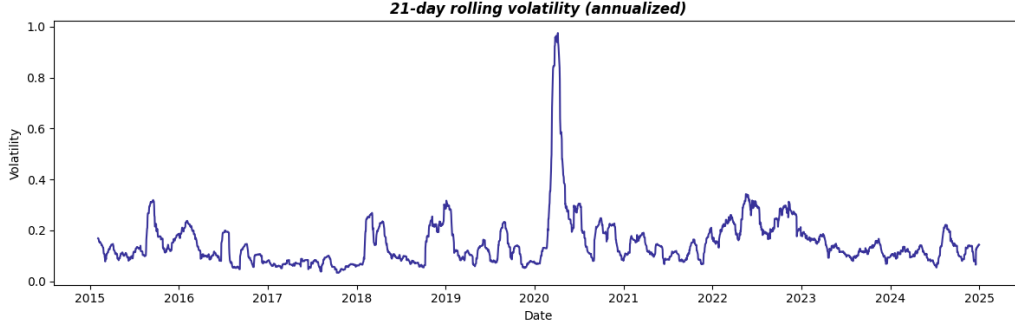


Figure 2.3: Twenty-one-day rolling annualized volatility of daily log returns.

2.3 Gaussian Hidden Markov specification

The empirical regime classification is based on a univariate Gaussian Hidden Markov Model estimated on daily log returns. Let $X_t \in \{1, \dots, m\}$ denote the latent regime at date t . Conditional on $X_t = i$, the return is assumed to follow

$$R_t \mid X_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad i = 1, \dots, m. \quad (2.3)$$

The latent state itself follows a first-order Markov chain with transition probabilities

$$p_{ij} = \Pr(X_t = j \mid X_{t-1} = i), \quad (2.4)$$

which are collected in the transition matrix $P = (p_{ij})$.

The HMM produces three types of outputs that are central for the empirical procedure. First, it estimates state-dependent means and standard deviations. Second, it estimates the transition matrix P . Third, it generates posterior state probabilities $\Pr(X_t = i \mid R_{1:T})$, which can be used both for classification and for assessing the sharpness of the inferred regime allocation.

2.4 Estimation strategy and parameter counting

Fitting the two-state and three-state Gaussian HMMs in the early stage is important since the initial fit will determine further pricing investigation. Fitting of two-state models is more interpretable as they make a simple classification of two regimes, while fitting the three-state model allows for an intermediate regime, but it can restrict outlier cases into one group only. In order to conduct fair comparisons, both models are fitted from different starting values: 20 start points are used with maximum iterations of 500 and convergence tolerance 10^{-8} .

A useful technical point is the number of free parameters. For a univariate Gaussian HMM with m states, the parameter count is

$$k(m) = (m - 1) + m(m - 1) + m + m = m^2 + 2m - 1. \quad (2.5)$$

The four terms correspond respectively to the initial state probabilities, the transition matrix, the state-dependent means, and the state-dependent variances. It follows that the two-state model has $k(2) = 7$ free parameters, while the three-state model has $k(3) = 14$. This parameter count enters directly into the information criteria used for model comparison.

2.5 Model comparison criteria and regime labelling

The two specifications are compared using several complementary diagnostics. The first is the maximized log-likelihood. Higher values indicate a better in-sample fit. Since log-likelihood alone always weakly rewards additional flexibility, the analysis also considers Akaike's Information Criterion and the Bayesian Information Criterion,

$$\text{AIC} = 2k - 2\ell, \quad \text{BIC} = k \log T - 2\ell, \quad (2.6)$$

where k is the number of free parameters, ℓ is the maximized log-likelihood, and T is the sample size. Lower values of AIC and BIC indicate a better trade-off between fit and parsimony.

A second diagnostic is posterior classification confidence. Let

$$\gamma_t(i) = \Pr(X_t = i \mid R_{1:T}) \quad (2.7)$$

denote the smoothed posterior probability of state i at date t . Average posterior confidence is computed as

$$\bar{c} = \frac{1}{T} \sum_{t=1}^T \max_i \gamma_t(i). \quad (2.8)$$

Values closer to one indicate that the model assigns observations to states more decisively. This criterion is useful because a model can fit well in likelihood terms while still producing diffuse or unstable classifications.

After the estimation process is complete, ranking states on the basis of increasing volatility can help reduce label switching. With two states, the lower-volatility state will be called the low-volatility state while the higher-volatility state will be labeled as the high-volatility state with respect to their respective dispersions and not mean returns. In case of three states, the third state will fall in between the two states and be identified as the medium-volatility state.

Chapter 3

Empirical Results

3.1 Comparison of the two-state and three-state models

Both the two-state and three-state Gaussian HMMs are estimated on the S&P 500 daily log-return series and assessed according to the criteria discussed in Chapter 2. Table 3.1 shows some main model summary statistics.

Table 3.1: Comparison of Gaussian HMM specifications.

States	Log-likelihood	AIC	BIC	Avg. posterior confidence	Smallest regime share
2	8191.6434	-16369.2869	-16328.5270	0.9512	0.2371
3	8301.2547	-16574.5094	-16492.9896	0.9533	0.0124

Firstly, the three-state model reaches a larger maximized log-likelihood, going up from 8191.6434 to 8301.2547. Taking into account a large difference compared to the increase in the number of parameters, the Akaike information criterion (AIC) and the Bayesian information criterion (BIC) for the three-state model are also lower. Thus, in terms of in-sample statistical fit, the three-state specification performs better.

However, the application of the model is not only about classifying observations into different regimes, but about creating a regime-switching stable and economic interpretable pricing mechanism. In this light, the three-state specification is less suitable for the pricing application. The growth in average posterior confidence is quite small, from 0.9512 to 0.9533. Moreover, the lowest share of a regime decreases drastically, going down from 23.71% to 1.24%. This means that the new state describes an extremely small proportion of observations, which is not enough to represent a certain state.

This issue is especially relevant for the following pricing application. A regime used in the regime-switching pricing model must be sufficiently persistent and economically meaningful. Thus, incorporating a state represented by only 31 observations can lead not only to higher log-likelihood but also to instability in volatility estimation and regime switch probability dynamics. Therefore, the three-state model is used here as a diagnostic tool for identifying periods of extreme market stress but not as the main pricing specification.

For all the reasons listed above, the two-state model is adopted further in the analysis as the operational specification. It provides a meaningful decomposition into high-volatility and low-volatility regimes, does not involve an extremely small tail regime, and creates a relatively stable transition structure between regimes in continuous time. However, the results from the three-state model will also be presented below, demonstrating that they do not depend critically on its assumption of intermediate or extreme volatility states. Further pricing results will be based on the two-state model.

3.2 The two-state classification

The two-state specification leads to a significant separation into a high-volatility and low-volatility states. Table 3.2 presents the state summary statistics under re-labeling by increasing volatility.

Table 3.2: State summary for the two-state Gaussian HMM.

State	Regime label	Mean daily return	Daily volatility	Annualized volatility	Count	Sample share	Expected duration (days)
0	Low-volatility	0.001007	0.007143	0.113388	1905	0.762915	74.8235
1	High-volatility	-0.001562	0.020511	0.325608	592	0.237085	21.9143

This implies that the low volatility regime covers 76.29% of the sample with an annualized volatility of 11.34%. The high volatility regime covers 23.71% with an annualized volatility of 32.56%. There is thus considerable gap between the two states in terms of their respective volatilities. On the other hand, the mean differences across states is of little information content.

Expected durations, calculated from $1/(1 - p_{ii})$, also lend themselves well to the idea of regimes. The duration of the low volatility regime lasts on average around 74.8 days, while the high volatility regime lasts for around 21.9 days. Calm periods are therefore longer and more prevalent, while turbulent periods are shorter but still persistent.

Both figures should be interpreted in conjunction as the first figure shows how the regimes map to the return data, while the second one shows the corresponding posterior probabilities, and thus indicates whether there is a clustering effect.

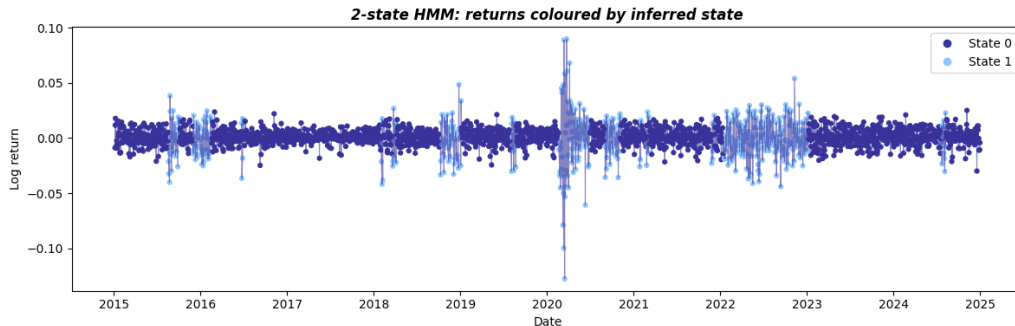


Figure 3.1: Daily log returns coloured by the inferred state in the two-state Gaussian HMM.

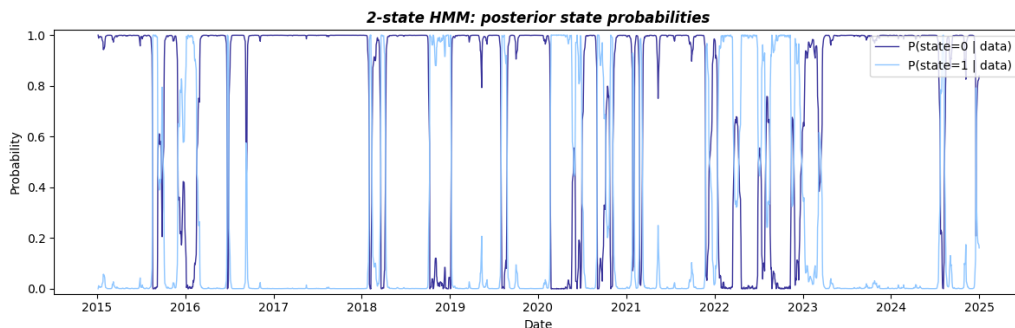


Figure 3.2: Posterior state probabilities under the two-state Gaussian HMM.

3.3 The three-state classification

The three-state model divides the turbulent part of the data set into intermediate and extreme states. State Summary in Table 3.3, sorted in order of increasing volatility, is presented

below.

Table 3.3: State summary for the three-state Gaussian HMM.

State	Regime label	Mean daily return	Daily volatility	Annualized volatility	Count	Sample share	Expected duration (days)
0	Low-volatility	0.001032	0.006659	0.105713	1741	0.697237	76.6486
1	Medium-volatility	-0.000619	0.015067	0.239186	725	0.290348	33.9140
2	High-volatility	-0.006119	0.052941	0.840410	31	0.012415	30.6050

Although it is the rarest state, the high volatility state exhibits a sharp spike in volatility with annualized volatility of 84.04% but occupies less than 1.24% of all data points with just 31 occurrences. Around 70% of the time is tranquil in the low volatility regime where daily fluctuations average around 10.57% annually. The remaining data points correspond to moderate volatility regime occupying almost one-third of the total time period with annualized volatility of 23.92%.

One of the reasons why such grouping of volatility makes sense is the consistency with observed financial-market behavior. In tranquil periods, volatility is relatively low; during hesitant periods, volatility reaches intermediate levels. Sharp spikes in volatility correspond to turbulent regimes. Returns usually decrease as the volatility increases. However, drawing strong conclusions based on the data might be quite erroneous since the estimates of averages are relatively imprecise, especially for the case of high volatility regime with just 31 occurrences. The empirical estimates support this interpretation. For instance, when looking at the state-coloured return plot, the third regime appears primarily in periods of maximum return magnitude. From the posterior-probability diagram, it can be seen that the high-volatility state comes and goes, appearing intermittently. Most of the time, there is movement between lower and moderate activities.

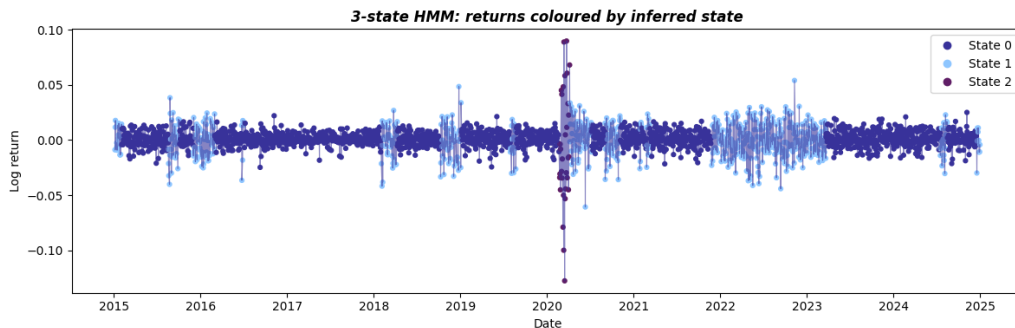


Figure 3.3: Daily log returns coloured by the inferred state in the three-state Gaussian HMM.

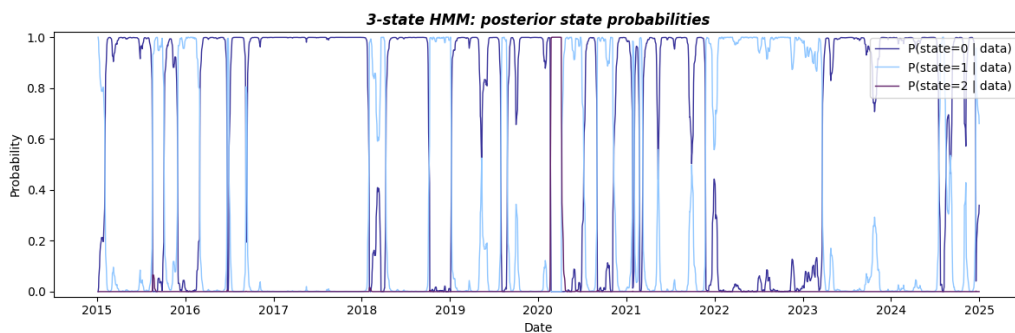


Figure 3.4: Posterior state probabilities under the three-state Gaussian HMM.

The three-state framework is useful because it distinguishes between normal times, intermediate periods, and stress periods. On the other hand, its high volatility regime is too small.

This means that it is useful for describing the data but less relevant as the base process for pricing options due to its instability. A two-state approach will do well in this case as it captures the key distinctions while avoiding a risky state regime.

3.4 Transition matrices and regime persistence

The HMM estimates also provide the discrete transition matrix P , which is central for interpreting regime persistence. In discrete time, the expected duration of regime i is

$$\mathbb{E}[D_i] = \frac{1}{1 - p_{ii}}. \quad (3.1)$$

For the two-state model, the transition matrix is shown in Table 3.4.

Table 3.4: Estimated transition matrix for the two-state Gaussian HMM.

	Low-volatility	High-volatility
Low-volatility	0.986635	0.013365
High-volatility	0.045632	0.954368

Diagonal elements being high suggest stability within regimes. With a probability of 0.986635, days characterized by a quiet market are expected to follow quiet market days. Days characterized by turbulent activities are expected to follow turbulent activity on the next day with a probability of 0.954368. It is easier to leave the turbulent state than the quiet state. It is also shorter as expected from the previous time-based analysis.

For the three-state model, the transition matrix is reported in Table 3.5.

Table 3.5: Estimated transition matrix for the three-state Gaussian HMM.

	Low-volatility	Medium-volatility	High-volatility
Low-volatility	0.986953	0.012386	0.000660
Medium-volatility	0.029479	0.970514	0.000008
High-volatility	0.000000	0.032674	0.967326

As indicated before, the diagonal entries are very high. Another important characteristic is the absence of direct movement between the two extreme state regimes. However, there is a presence of indirect transition through the intermediary regime making it an excellent economic explanation. This explains why this sort of specification may pose challenges for practical use in option pricing.

3.5 Continuous-time approximation of switching intensities

The estimated daily transition matrices can be converted into approximate continuous-time generators using

$$Q \approx \frac{1}{\Delta t} \log(P), \quad (3.2)$$

where $\Delta t = 1/252$ years.

For the two-state model, the generator approximation is reported in Table 3.6.

Table 3.6: Continuous-time generator approximation for the two-state model.

	Low-volatility	High-volatility
Low-volatility	-3.471362	3.471362
High-volatility	11.852538	-11.852538

Off-diagonal entries represent the annualized intensity of transition between the two state regimes. Consistent with the previous finding, the transition from the high volatility regime to the low volatility regime is greater compared to vice versa because of the shorter nature of high volatility regimes.

For the three-state model, the generator approximation is reported in Table 3.7.

Table 3.7: Continuous-time generator approximation for the three-state model.

	Low-volatility	Medium-volatility	High-volatility
Low-volatility	-3.357081	3.186777	0.170304
Medium-volatility	7.591213	-7.590586	-0.000627
High-volatility	-0.127717	8.499169	-8.371452

For the case of the three-state regime, most of the transitions are between adjacent state regimes. Low volatility regimes mostly tend to transition into the intermediary state regime while high volatility regimes tend to transition into the intermediary state regimes.

The rows sum approximately to zero, as required for a continuous-time generator matrix. However, a valid generator should have non-negative off-diagonal entries, since these entries represent transition intensities. The small negative off-diagonal values in the three-state approximation should therefore be interpreted as numerical artefacts of the matrix-logarithm approximation, likely related to the limited number of observations in the extreme high-volatility regime. This reinforces the view that the three-state model is useful descriptively, but less stable as an input for the pricing application.

Chapter 4

Option Pricing Application

4.1 Pricing exercise and walk-forward design

Thus far, the evidence on volatility regimes has been clear from the analysis of S&P 500 returns data. The current section employs volatility regimes to price near-term index call options. The aim is not to fully calibrate the market, but to examine whether the historical estimate of regime switching dynamics leads to reasonable option prices.

The pricing process is applied to an almost at-the-money 30-day S&P 500 call option. At each observation day t , empirical market prices are contrasted against prices of a two-state regime-switching Black-Scholes model. Two regimes are estimated using a Gaussian Hidden Markov Model, while the associated probabilities of transitioning between the regimes as well as the regime probabilities predicted by the model are used for pricing.

To prevent look-ahead bias, HMM is estimated in a walk-forward fashion. At the start of each calendar month, HMM is estimated using the continuously expanded window of S&P 500 daily log return observations until the last day of the previous month. States are sorted based on their levels of volatility in ascending order, whereby the first state corresponds to the low-volatility regime and the second state to the high-volatility regime.

Within each pricing month, HMM parameters remain constant. Probabilities of regimes change in a sequential manner with the arrival of additional returns. For pricing on date t , the model uses the one-step-ahead predicted probability

$$\tilde{p}_t = p_{t-1}^{filt} P, \quad (4.1)$$

where p_{t-1}^{filt} is the filtered regime probability after observing returns up to date $t - 1$, and P is the estimated daily transition matrix. This gives

$$\tilde{p}_t = \left(p_{L,t}^{pred}, p_{H,t}^{pred} \right), \quad (4.2)$$

where $p_{L,t}^{pred}$ and $p_{H,t}^{pred}$ are the probabilities assigned to the low- and high-volatility regimes before observing the return on date t .

After the return on date t is observed, the predicted probability is updated through Bayes' rule. If the regime-specific return densities are denoted by $f_L(R_t)$ and $f_H(R_t)$, the filtered probability is

$$p_{i,t}^{filt} = \frac{p_{i,t}^{pred} f_i(R_t)}{p_{L,t}^{pred} f_L(R_t) + p_{H,t}^{pred} f_H(R_t)}, \quad i \in \{L, H\}. \quad (4.3)$$

The filtered probability is then used as the input for the next day's prediction. In this way, the model distinguishes between the probability available before observing the pricing-date return and the probability updated after that return is known.

This timing convention is important. The option price on date t is computed using $p_{L,t}^{pred}$ and $p_{H,t}^{pred}$, not $p_{L,t}^{filt}$ and $p_{H,t}^{filt}$. The latter incorporate information from the same-day return and

are therefore used only to update the next day's regime forecast.

4.2 Walk-forward regime inputs

Between the years 2020 and 2024, the regime signal for each day is obtained using a hidden Markov model (HMM) analysis performed by a walk forward approach. Even though pricing starts from 1 January 2020, the returns observations go back up to 6 January 2015, with even earlier observations going back almost five years from there. This leaves sufficient historical data before the first pricing date to estimate the initial HMM.

For each pricing month, the estimated HMM provides the following quantities:

- the annualized low-volatility regime parameter σ_L ;
- the annualized high-volatility regime parameter σ_H ;
- the daily transition matrix P ;
- the continuous-time switching intensities q_{LH} and q_{HL} ;
- the one-step-ahead predicted probabilities $p_{L,t}^{pred}$ and $p_{H,t}^{pred}$.

The discrete transition matrix is converted into a continuous-time generator using

$$Q \approx 252 \log(P), \quad (4.4)$$

where 252 is the number of trading days in a year. In the two-state case,

$$Q = \begin{pmatrix} -q_{LH} & q_{LH} \\ q_{HL} & -q_{HL} \end{pmatrix}. \quad (4.5)$$

The values lying outside the diagonal become the measure of annual switch rates for both the states. The model does not have any time gap as such, and therefore, these rates are linked between the daily Hidden Markov Model calculations and dynamically changing equations.

Table 4.1 reports summary statistics for the main walk-forward regime inputs.

Table 4.1: Summary of walk-forward regime inputs, 2020–2024.

	σ_L	σ_H	p_L^{pred}	p_H^{pred}	p_L^{filt}	p_H^{filt}	q_{LH}	q_{HL}
Mean	0.1116	0.3571	0.7323	0.2677	0.7276	0.2724	3.6268	14.0094
Minimum	0.1044	0.2604	0.0403	0.0155	0.0000	0.0043	2.5602	9.8963
25th percentile	0.1107	0.3287	0.4689	0.0236	0.4451	0.0101	3.3285	11.2075
Median	0.1115	0.3429	0.9502	0.0498	0.9611	0.0389	3.7907	12.6038
75th percentile	0.1131	0.3948	0.9764	0.5311	0.9899	0.5549	4.0052	17.1669
Maximum	0.1145	0.4444	0.9845	0.9597	0.9957	1.0000	4.2024	19.7420
Standard deviation	0.0021	0.0415	0.3379	0.3379	0.3624	0.3624	0.4448	3.2584

The expected low-volatility regime is extremely persistent, having on average an annualized volatility level equal to 11.16%, ranging between 10.44% and 11.45%. On the other hand, the expected high-volatility regime experiences a higher degree of volatility fluctuations, as it has an annualized mean volatility level of 35.71% ranging between 26.04% and 44.44%.

Indeed, the above results reflect the interpretation that the low volatility regime is a persistent state, whereas the high-volatility one is a stress regime.

The expected probability of high volatility shows a mean of 26.77% and a median of 4.98%. From the above values, we conclude that the model assigns low probabilities to the high-volatility regime, yet at times, it experiences a significant increase in assigning such a probability, especially during turbulence regimes. At most, the probability assigned by the model to high volatility amounts to 95.97%.

As for the switching intensities, the mean intensity of switching from the low volatility regime to the high volatility regime is $q_{LH} = 3.6268$, whereas the mean intensity of switching from the high volatility regime to the low volatility regime is $q_{HL} = 14.0094$. As $q_{HL} > q_{LH}$, the model infers that the high-volatility regime is less persistent than the low-volatility regime.

4.3 Regime-switching Black–Scholes pricing equation

The pricing model is based on a two-state regime-switching Black–Scholes system. Let $V_L(S, t)$ denote the option value conditional on the current regime being low volatility, and let $V_H(S, t)$ denote the value conditional on the current regime being high volatility. The coupled pricing equations are

$$\frac{\partial V_L}{\partial t} + \frac{1}{2}\sigma_L^2 S^2 \frac{\partial^2 V_L}{\partial S^2} + rS \frac{\partial V_L}{\partial S} - rV_L + q_{LH}(V_H - V_L) = 0, \quad (4.6)$$

and

$$\frac{\partial V_H}{\partial t} + \frac{1}{2}\sigma_H^2 S^2 \frac{\partial^2 V_H}{\partial S^2} + rS \frac{\partial V_H}{\partial S} - rV_H + q_{HL}(V_L - V_H) = 0. \quad (4.7)$$

The terminal payoff is common across regimes. For a European call,

$$V_L(S, T) = V_H(S, T) = \max(S - K, 0). \quad (4.8)$$

For a European put,

$$V_L(S, T) = V_H(S, T) = \max(K - S, 0). \quad (4.9)$$

The coupling terms $q_{LH}(V_H - V_L)$ and $q_{HL}(V_L - V_H)$ distinguish the model from a simple weighted Black–Scholes approximation. The value in the low-volatility regime depends on the possibility of switching to the high-volatility regime before maturity, and the value in the high-volatility regime depends on the possibility of switching back to the low-volatility regime. Therefore, the regime-conditional prices already incorporate possible future regime switches. Once the two conditional prices have been obtained, the final regime-switching price is

$$C_t^{RS} = p_{L,t}^{pred} V_L(S_t, t) + p_{H,t}^{pred} V_H(S_t, t). \quad (4.10)$$

This aggregation works because the current regime is latent. The observed price generated by the model is the expectation of the regime-conditional values under the predicted regime distribution:

$$C_t^{RS} = \mathbb{E} [V_{X_t}(S_t, t) \mid \mathcal{F}_{t-1}]. \quad (4.11)$$

In the two-state case, this expectation is exactly the probability-weighted average above.

4.4 Numerical solution

A solution in closed form, common under the classic Black-Scholes framework, typically eludes the coupled pricing mechanism. Because of this divergence, numerical computation steps in, employing a fully implicit finite-difference approach, to determine option values.

The stock-price domain is truncated to a finite interval $[0, S_{\max}]$ and discretized into a grid. At maturity, the option value is initialized at the payoff. The solution then proceeds backward in calendar time, equivalently forward in time-to-maturity $\tau = T - t$. At each time step, the first and second derivatives with respect to the underlying price are approximated by finite differences.

For each regime, the finite-difference approximation produces a tridiagonal operator. Because the two regimes are linked by switching intensities, the low- and high-volatility values are solved jointly through a block linear system of the form

$$\begin{pmatrix} A_L & -\Delta t q_{LH} I \\ -\Delta t q_{HL} I & A_H \end{pmatrix} \begin{pmatrix} V_L^{new} \\ V_H^{new} \end{pmatrix} = \begin{pmatrix} V_L^{old} \\ V_H^{old} \end{pmatrix}. \quad (4.12)$$

The off-diagonal blocks contain the regime-switching terms. This joint solution is essential: solving the two Black-Scholes equations separately would ignore the possibility that volatility changes before expiration.

Boundary conditions are imposed at $S = 0$ and $S = S_{\max}$. For a call option,

$$C(0, \tau) = 0, \quad C(S_{\max}, \tau) \approx S_{\max} - Ke^{-r\tau}. \quad (4.13)$$

For a put option,

$$P(0, \tau) = Ke^{-r\tau}, \quad P(S_{\max}, \tau) \approx 0. \quad (4.14)$$

Since only volatility switches in the present specification and the risk-free rate is common across states, the same boundary values are used for both regimes.

4.5 Option data and pricing assumptions

For every trading day, the ATM dataset contains one observed SPX call option price together with the corresponding index level, strike, expiration date, and short-term interest rate. The current implementation uses the observed strike and the observed expiration date. Therefore, the pricing exercise no longer relies on the older simplifying approximation $K_t = S_t$ or on a fixed maturity $T = 30/365$.

For each date t , the inputs are:

- S_t , the S&P 500 index level;
- K_t , the observed strike price;
- T_t , the time to maturity computed from the observed expiration date;
- r_t , the 3-month Treasury-bill rate used as the risk-free rate proxy;
- C_t^{mkt} , the observed market option price;
- $\sigma_L, \sigma_H, q_{LH}, q_{HL}, p_{L,t}^{pred}$ and $p_{H,t}^{pred}$ from the walk-forward HMM.

The time to maturity is computed as

$$T_t = \frac{\text{expiry_date}_t - \text{date}_t}{365}. \quad (4.15)$$

Rows where the maturity value is non-positive, the strike is not available, the price is not available, or the market price is non-positive are dropped from the sample used to price the options. The reason for this is that we want to remain faithful to the data in our analysis and not introduce extra error by enforcing that all options be exactly at the money or exactly 30 days away from maturity.

The significance of this decision relates to the interpretation of the results below. While we denote the option as being at the money (ATM), considering the nature of the data, the model option price is calculated using the strike and maturity as actually observed from each row.

4.6 Pricing results

For each date, the regime-switching PDE produces two conditional prices, V_L and V_H , and the final model price C_t^{RS} is obtained using the predicted regime probabilities.

Table 4.2: Pricing error summary for ATM 30-day SPX call options.

Metric	Observations	MAE	RMSE	Success rate
Value	1249	15.2067	21.1527	1.0000
Metric	Mean relative error	Median relative error	Mean market price	Mean model price
Value	0.1801	0.1570	83.9136	86.0268

The average absolute pricing error is 15.21 index points, while the root mean squared error is 21.15. The mean relative error is 18.01%, and the median relative error is 15.70%. The mean model price, 86.03, is slightly above the mean market price, 83.91, suggesting a modest average overpricing bias over the full period. The success rate is 100%, indicating that the numerical PDE procedure generated valid prices for all observations in the pricing sample.

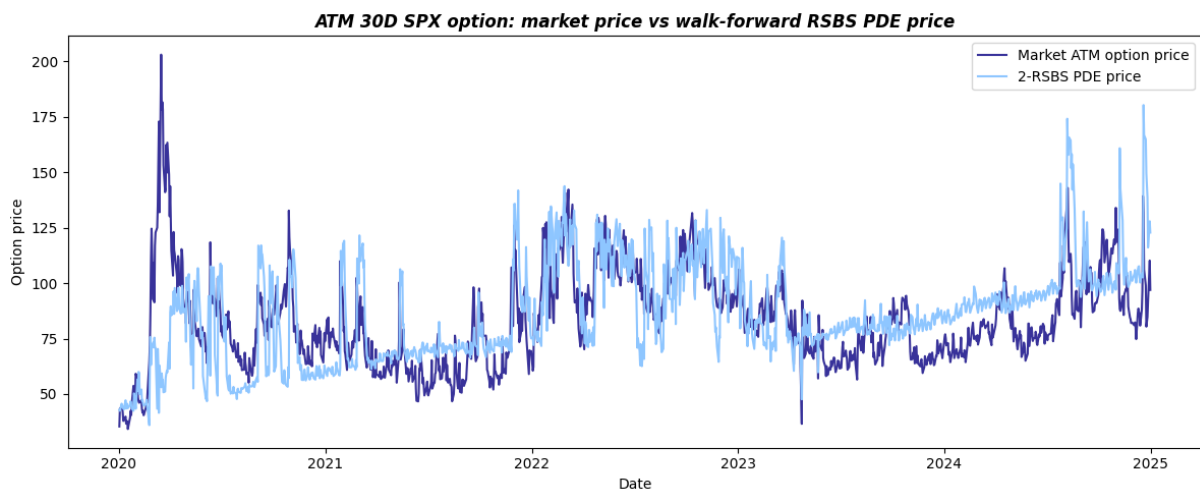


Figure 4.1: ATM 30-day SPX option market price versus walk-forward regime-switching Black–Scholes price.

As seen from Figure 4.1, market prices are compared to those produced by the regime-switching Black-Scholes walk-forward model. Although there are cases where the two series move similarly in one direction, especially during prolonged stretches of high volatility, noticeable differences are found in the extent of the peaks reached. Rapid jumps in actual prices tend to outperform those predicted by the model, for example at the start of 2020 where actual levels increased faster and higher compared to the modeled prices. These divergences highlight shortcomings in the model's ability to replicate sudden regime changes.

The mismatch is economically meaningful. Market prices are forward-looking and incorporate risk-neutral expectations, volatility risk premia, skew, demand for crash insurance, and liquidity factors. On the other hand, regime switching models use historical returns that are assessed under the real probability measure. Thus, it would be more sensible to consider it as a historical benchmark, rather than a valuation model.

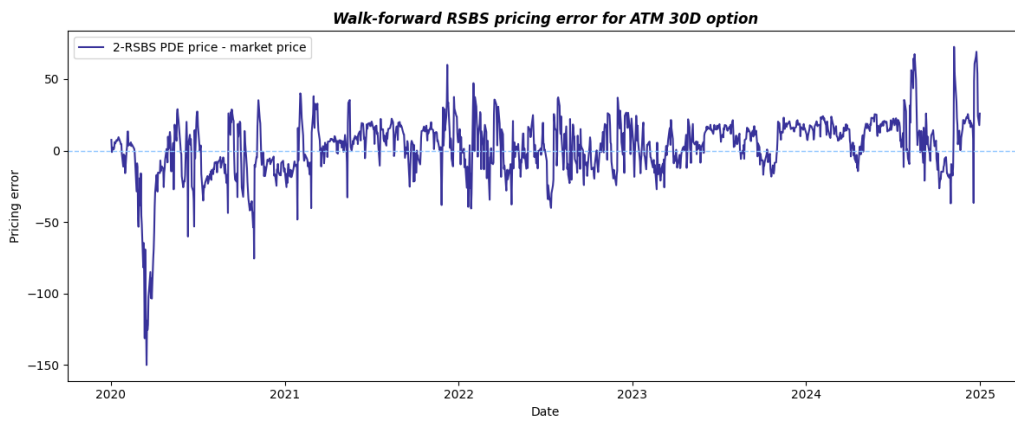


Figure 4.2: Pricing error of the walk-forward regime-switching Black–Scholes model.

Figure 4.2 shows the pricing error

$$Error_t = C_t^{RS} - C_t^{mkt}. \quad (4.16)$$

If above zero, it is an indication of overestimation of the model. If below zero, on the other hand, then it is an indicator of underestimation. During the year of increased volatility, deviations become very low and go into the negative zone, showing significant undervaluation of implied volatility. Such deviation occurs since the shift depends on past performance but is unable to adjust as quickly as market-implied volatility to changes in investors' expectations. Although correlated with the past, the mechanism cannot catch immediate valuation change caused by market expectations.

During 2021 until 2022, deviations stay near zero while being unstable. Starting from 2023 and into 2024, predictions tend to be larger than observed market prices, which could happen because there was less volatility in market conditions compared to the turbulent period. The deviations towards the end become wider, which is evidence of difficulty in adapting to sudden volatility changes in a monthly updated HMM approach.

For the walk-forward option pricing application, the one-step-ahead predicted regime probabilities of HMM will be used. They are responsible for determining the starting distribution for the regimes in the regime-switching Black–Scholes partial differential equation. Thus, the prediction uses only data from the previous month to avoid look-ahead bias. With high predicted volatility regimes weighted higher, the value of options will be more expensive.

4.7 Comparison with a Black–Scholes benchmark

To assess whether the regime-switching structure improves pricing performance, the regime-switching Black–Scholes model is compared with a standard Black–Scholes benchmark. The benchmark uses the same option inputs as before, namely S_t , the observed strike K_t , the observed time to maturity T_t , and r_t , but replaces the two-regime volatility structure with a single rolling historical volatility estimate.

To avoid look-ahead bias, the rolling volatility used on date t is computed using returns available up to date $t - 1$:

$$\sigma_t^{roll} = \text{Std}(R_{t-w}, \dots, R_{t-1}) \sqrt{252}. \quad (4.17)$$

The baseline window is set to $w = 21$ trading days, approximately one trading month. The benchmark price is denoted by C_t^{BS} and is compared with the regime-switching price C_t^{RS} and the observed market price C_t^{mkt} .

Table 4.3 reports the main pricing performance statistics for the two models.

Table 4.3: Pricing performance comparison between RSBS and rolling-volatility Black–Scholes.

Metric	Regime-switching BS	Rolling-volatility BS
MAE	15.2067	17.6607
RMSE	21.1527	26.9039
Mean error	2.1133	-0.5351
Median error	4.8724	-2.6334
Mean relative error	0.1801	0.2031
Median relative error	0.1570	0.1560
Mean model price	86.0268	83.3785
Mean market price	83.9136	83.9136

The regime switching approach outperforms the rolling volatility Black Scholes model across all the considered error measures. In particular, MAE is reduced from 17.66 to 15.21, while RMSE drops from 26.90 to 21.15. In addition, MAPE decreases from 20.31% to 18.01%. Therefore, it can be inferred that incorporating regimes of persistent low and high volatility in order to capture the underlying volatility dynamics adds value for option price discovery beyond rolling historical volatility estimates.

Besides the above-mentioned descriptive statistics describing option pricing performance, Table 4.4 summarizes a more direct comparison of the models through loss differentials. The loss difference between the models is defined by taking the difference between RSBS and the rolling volatility Black Scholes models' losses, which takes negative values when the regime-switching approach performs better. Both an economic significance analysis and statistical tests on absolute and squared loss differentials have been included in Table 4.4. Because option-pricing errors can be serially correlated, especially when the same model is updated through time, the Diebold–Mariano statistics are also reported with a HAC correction.

Table 4.4: Economic and statistical comparison between RSBS and rolling-volatility Black–Scholes.

Comparison	Loss measure	Obs.	RSBS	BS	BS – RSBS	Improvement	Test stat.	<i>p</i> -value
Economic magnitude	MAE	1249	15.2067	17.6607	2.4540	13.90%	–	–
Economic magnitude	RMSE	1249	21.1527	26.9039	5.7512	21.38%	–	–
Paired <i>t</i> -test	$ e^{RS} - e^{BS} $	1249	15.2067	17.6607	2.4540	13.90%	-4.1182	0.000041
Diebold–Mariano HAC	Absolute error	1249	15.2067	17.6607	2.4540	13.90%	-1.7360	0.082811
Diebold–Mariano HAC	Squared error	1249	21.1527	26.9039	5.7512	21.38%	-1.4629	0.143757

The paired *t*-test strongly rejects equality of mean absolute errors, with the one-sided evidence supporting RSBS even more directly. The HAC Diebold–Mariano results are more conservative: the absolute-error statistic is significant at the 10% level in the two-sided test and at the 5% level in the one-sided test, while the squared-error statistic is weaker. This distinction is useful. It suggests that the regime-switching model produces an economically visible reduction in pricing errors, but that the statistical strength of this gain depends on how serial dependence and extreme observations are treated. Therefore, the result should be read as supportive evidence for RSBS rather than as a claim of complete dominance in every loss function.

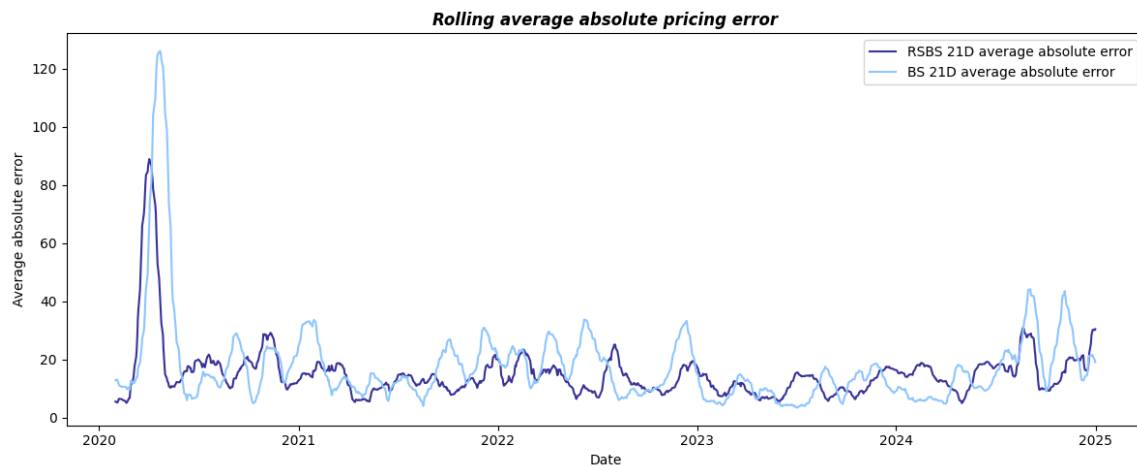


Figure 4.3: Twenty-one-day rolling average absolute pricing error for RSBS and rolling-volatility Black–Scholes.

The rolling average absolute error in Figure 4.3 implies that the RSBS model is generally more stable compared to its rolling-volatility counterpart. This observation holds particularly true for the period characterized by the most significant volatility shock, at which point the Black–Scholes benchmark shows an extremely strong reaction to changes in realized volatility, resulting in large errors.

It may be suggested that the regime-switching approach offers a more intuitive representation of historical volatility patterns. While the exact gain differs across time periods and conditions, overall performance shows improvement. Lower MAE, lower RMSE, and lower average relative deviation all imply an informational edge. HMM models appear better at estimating prices in comparison to rolling input variables.

Table 4.5 explores whether or not the price conclusions are affected by the amount of historical information used to train the HMM in the walk-forward experiment. The default setting includes five years of information to estimate the hidden Markov model. Rolling windows react faster to market changes but generate less accurate regime states and transition matrices.

Table 4.5: Robustness of RSBS pricing results to the HMM rolling estimation window.

Rolling window	Obs.	Failed months	MAE	RMSE	Mean error	Median error	Mean relative error	Median relative error	Mean model price
5 years	1748	0	17.4759	23.4834	9.1167	8.9587	0.2483	0.2064	80.8682
3 years	1748	0	21.2593	27.8295	14.2379	12.3175	0.3047	0.2462	85.9894
2 years	1748	0	22.5439	29.0537	14.3787	12.5107	0.3347	0.2702	86.1302

It is advantageous to use the estimation window that is extended over five years. If the estimation window of HMM decreases from five years to either three years or two years, it causes an increase in the mean absolute error (MAE) and root mean squared error (RMSE). It also makes the model price significantly higher than the market price. The results indicate that too short an estimation window period can make the latent regime inputs sensitive but does not imply an improvement in accuracy for valuing options.

4.8 Implied-volatility comparison

The previous diagnostics compare models in price space. As an additional check, the market price and the RSBS model price are converted into Black–Scholes implied volatilities. For each ATM date t , the inversion uses the same option inputs as the pricing exercise:

$$S_t, \quad K_t, \quad T_t, \quad r_t. \quad (4.18)$$

Here K_t is the observed strike and T_t is computed from the observed expiration date. The comparison therefore does not use the older approximation $K_t = S_t$ or a fixed maturity $T = 30/365$.

The two implied-volatility series are

$$\sigma_t^{mkt,IV} = BS^{-1} \left(C_t^{mkt}; S_t, K_t, T_t, r_t \right), \quad (4.19)$$

and

$$\sigma_t^{RSBS,IV} = BS^{-1} \left(C_t^{RSBS}; S_t, K_t, T_t, r_t \right). \quad (4.20)$$

The implied-volatility error is

$$IVError_t = \sigma_t^{RSBS,IV} - \sigma_t^{mkt,IV}. \quad (4.21)$$

A positive value means that the RSBS price embeds a higher Black–Scholes-equivalent volatility than the market price, while a negative value means that the RSBS price embeds too little volatility relative to the market.

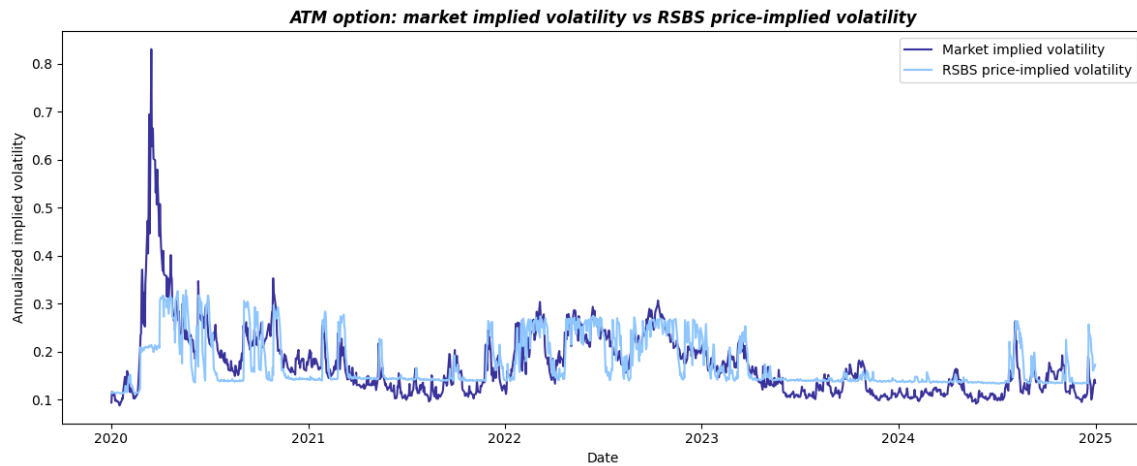


Figure 4.4: ATM option market implied volatility versus RSBS price-implied volatility.

The advantage of this test is that it separates the variation in the price level from the inherent volatility in the price itself. The RSBS implied volatility tends to be smoother than market implied volatility, as it is based on HMM estimates from history updated monthly, whereas market implied volatility is more sensitive because it directly responds to the changes in the options market, risk premium, skew, liquidity, and forward expectations. Significant differences between the two can be expected during periods of market stress.

4.9 Interpretation and limitations

Results about pricing suggest that the regime-switching Black–Scholes model captures relevant volatility-regime effects. As the estimated probability of high-volatility regime increases, higher weights are assigned to the high-volatility option value, resulting in a higher price. This is consistent with conventional option pricing where call option prices increase along with volatility.

At the same time, agreement with observed option prices is partial. The greatest discrepancies can be observed during sudden periods of stress when the adjustment of implied volatility happens faster than the change of historical regimes. This mismatch stems from the gap between the use of historical HMM estimates based on actual probabilities and the consideration of risk-adjusted market prices.

Several simplifications are still present. The two-state volatility regime is used, constant parameters are considered within months, volatility skew and term structure of volatility, jumps, liquidity considerations, and changes in market expectations are not captured.

In conclusion, additional insight beyond the simple benchmark of historical volatility is provided by regimes of volatility. However, only historical regimes are not enough for market-consistent option pricing since one should consider either risk-neutral calibration or information from options.

Chapter 5

Out-of-the-money option pricing

While the previous analysis concentrated on pricing near at-the-money SPX calls, this section employs the pricing method described before on out-of-the-money call options. The dataset utilized for this particular analysis consists of one option per trading day, with the target moneyness of options

$$\frac{S_t}{K_t} \approx 0.90. \quad (5.1)$$

Given that we work with call options, it means that strike levels for out-of-the-money options are higher than the current index level.

The aim of this experiment is to see whether the regime-switching Black-Scholes pricing approach can be helpful when pricing options with high sensitivity to tail behavior and volatility expectations. Indeed, out-of-the-money options are much more influenced by volatility skew, crash risk premia, and protection demand from other investors, while none of these components are included in our regime-switching model based on historical data.

Again, the same walk-forward HMM regime estimates are employed as in the previous analysis. At each point in time, the model uses the same set of variables as inputs: the index level S_t , the strike K_t , the risk-free rate r_t , market price C_t^{mkt} , and all the regime-switching variables computed from HMM estimates. By default, in the exercise involving out-of-the-money options, the maturity is fixed at 30 calendar days; however, the experiment can also be run with observed expiration dates in the notebook. Again, the error of pricing is defined as

$$Error_t = C_t^{model} - C_t^{mkt}. \quad (5.2)$$

Table 5.1 reports the main comparison between the regime-switching Black–Scholes model and the rolling-volatility Black–Scholes benchmark.

Table 5.1: Pricing performance for OTM 90% SPX call options.

Model	Obs.	MAE	RMSE	Mean error	Median error	Mean relative error	Median relative error	Mean model price
Regime-switching BS	1249	7.9247	10.5414	6.9683	4.8135	14.4313	11.2684	8.9514
Rolling-volatility BS	1249	6.0703	20.5252	5.7235	0.2532	2.8544	0.9960	7.7066
				Mean market price		1.9831		

In the OTM case, there is considerable variation between the RSBS and RVBS models and the at-the-money mean. The former has a mean of 8.95, and the latter has a mean of 7.71, while the observed mean is only 1.98.

While the rolling-volatility method gives a smaller MAE than the regime-switching technique, 6.07 versus 7.92, the regime-switching model significantly outperforms the rolling-volatility approach with respect to RMSE. With a RMSE of 10.54 and 20.53, respectively, this means that RSBS is better at preventing extreme pricing mistakes.

Relative errors must be carefully considered, because many of the OTM option prices are close to 0 index points. In such cases, even minor pricing mistakes in index points can cause extreme percentage errors, as seen in Table 5.1.

The weaker OTM pricing performance is expected. The volatility models based on historic data fail to incorporate important factors related to OTM options pricing. These include the presence of volatility smile, risk-neutral skew, crash-risk premia, liquidity effects, and demand for protection. Consequently, the RSBS model should be thought of as a benchmark, and not as a stand-alone OTM pricing technique.

Figure 5.1 compares the observed OTM market option price with the regime-switching and rolling-volatility Black–Scholes model prices.

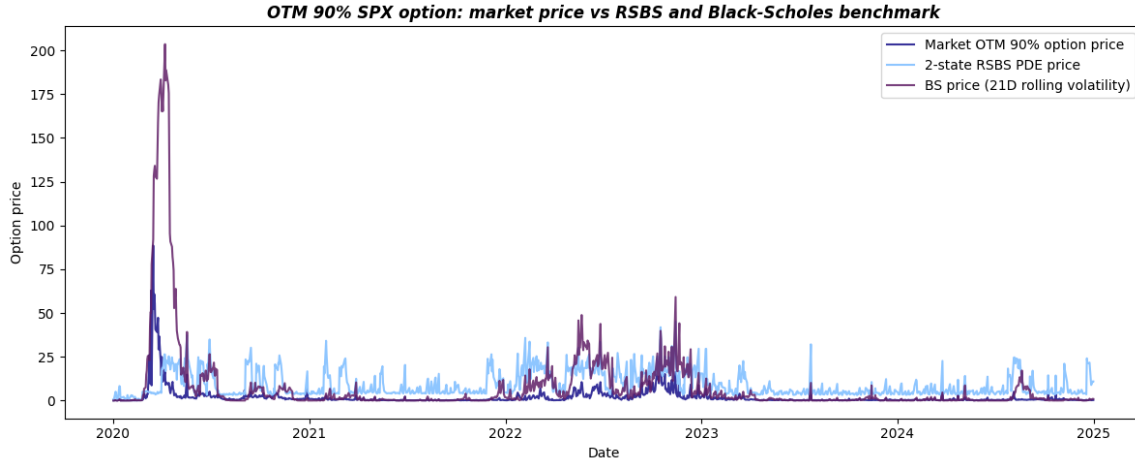


Figure 5.1: OTM 90% SPX option market price versus RSBS and rolling-volatility Black–Scholes prices.

5.1 OTM implied-volatility comparison

The final OTM diagnostic converts both the observed market price and the original RSBS model price into Black–Scholes implied volatilities. For each date t , the inversion uses the same option inputs as the OTM pricing exercise: S_t , K_t , T_t , and r_t .

In the baseline OTM specification, T_t is fixed at 30 calendar days. If the implementation switch is changed, the same diagnostic can instead use the maturity implied by the observed expiration date. The two implied-volatility series are

$$\sigma_t^{mkt,IV} = BS^{-1} \left(C_t^{mkt}; S_t, K_t, T_t, r_t \right), \quad \sigma_t^{RSBS,IV} = BS^{-1} \left(C_t^{RSBS}; S_t, K_t, T_t, r_t \right). \quad (5.3)$$

The implied-volatility error is

$$IVError_t^{OTM} = \sigma_t^{RSBS,IV} - \sigma_t^{mkt,IV}. \quad (5.4)$$

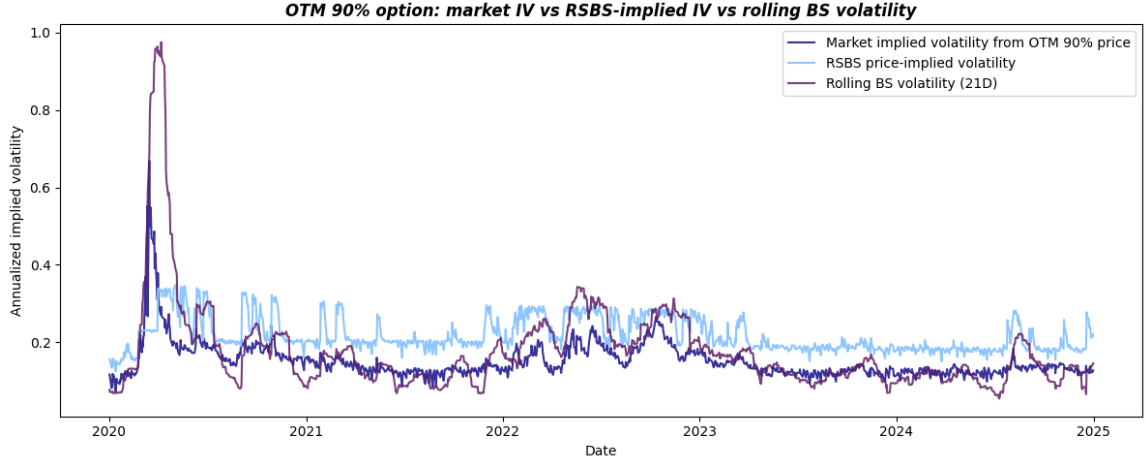


Figure 5.2: OTM 90% option market implied volatility, RSBS price-implied volatility, and rolling Black–Scholes volatility.

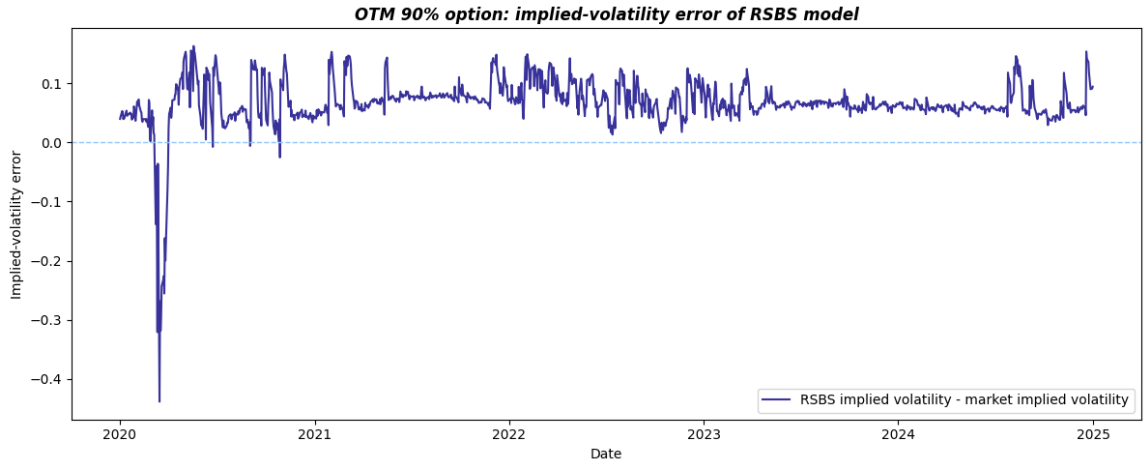


Figure 5.3: OTM 90% option implied-volatility error, $\sigma_t^{RSBS,IV} - \sigma_t^{mkt,IV}$.

This is a more reasonable comparison than comparing implied volatility from the market directly with the low and high volatilities generated from the HMM models. The low and high volatilities from the HMM model are based on physical measure estimations from the returns. On the other hand, both $\sigma_t^{mkt,IV}$ and $\sigma_t^{RSBS,IV}$ are based on the Black–Scholes equivalent volatilities, which are calibrated from option prices with similar strikes, maturities, and interest rate parameters.

5.2 Walk-forward OTM volatility adjustment

The implied volatility comparison helps to determine if the volatility embedded in the RSBS price is too high or too low compared to the market price. This can be examined directly using a very simple walk-forward volatility calibration. Specifically, for every month, the calibration parameter is calculated by the median ratio between market implied volatility and RSBS price-implied volatility from the last month:

$$\alpha_m = \text{median}_{t \in m-1} \left(\frac{\sigma_t^{mkt,IV}}{\sigma_t^{RSBS,IV}} \right). \quad (5.5)$$

The adjustment is then applied multiplicatively to all HMM regime volatilities during month m :

$$\sigma_{i,t}^{adj} = \alpha_m \sigma_{i,t}. \quad (5.6)$$

One month delay helps to avoid look-ahead bias. Clipping window used in the implementation process prevents the adjustment from being excessively volatile when it comes to noisy estimates of OTM implied-volatility.

Adjusted model is not claimed to be a completely calibrated option-pricing framework. Rather, it serves a purely diagnostic purpose. Specifically, adjustment procedure separates two effects from each other. The original RSBS model examines whether only regime-switching volatility can adequately describe price dynamics. The adjusted one studies whether simple lagged adjustment, based on information from options, can provide better pricing results without relying on market prices on the same day.

Table 5.2: OTM 90% option pricing performance: original RSBS, alpha-adjusted RSBS, and rolling-volatility Black–Scholes.

Model	MAE	RMSE	Mean err.	Median err.	Mean rel. err.	Median rel. err.
Original RSBS	7.9247	10.5414	6.9683	4.8135	14.4313	11.2684
Alpha-adj. RSBS	2.4082	6.3987	0.9371	0.6379	2.6555	1.8280
Rolling-vol. BS	6.0703	20.5252	5.7235	0.2532	2.8544	0.9960

Table 5.2 contains quantitative results of the OTM volatility adjustment exercise. Alpha-adjusted RSBS model leads to the smallest MAE. In particular, it brings MAE from 7.9247 in the original RSBS model down to 2.4082. Furthermore, the adjusted RSBS framework also lowers RMSE to 6.3987 (versus 10.5414 of the original framework). This means that volatility adjustment helps to both minimize average pricing errors and their variability. Moreover, volatility-adjusted RSBS model outperforms rolling-volatility Black-Scholes in terms of MAE and RMSE. However, it should be noted that rolling volatility Black-Scholes shows smaller median relative errors.

This comparison suggests that the primary problem in the OTM case of the original RSBS model is not the regime switching part, but rather its volatility level as defined using historical HMM estimates. Once regime-switching volatilities are adjusted using lagged option-implied data, pricing performance is improved greatly. Thus, this suggests that OTM option pricing requires information about risk-neutral level of volatility and volatility skew.

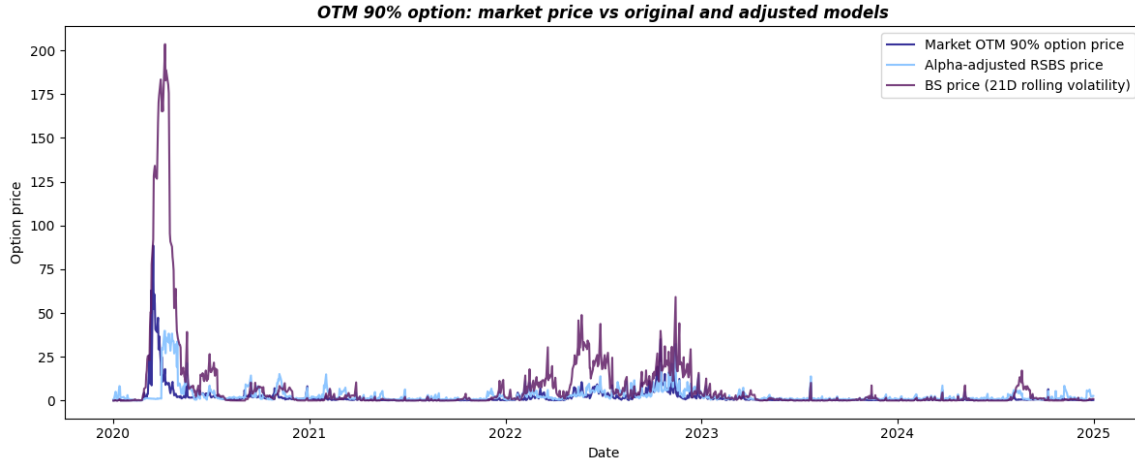


Figure 5.4: OTM 90% option market price versus alpha-adjusted RSBS and rolling-volatility Black–Scholes prices.

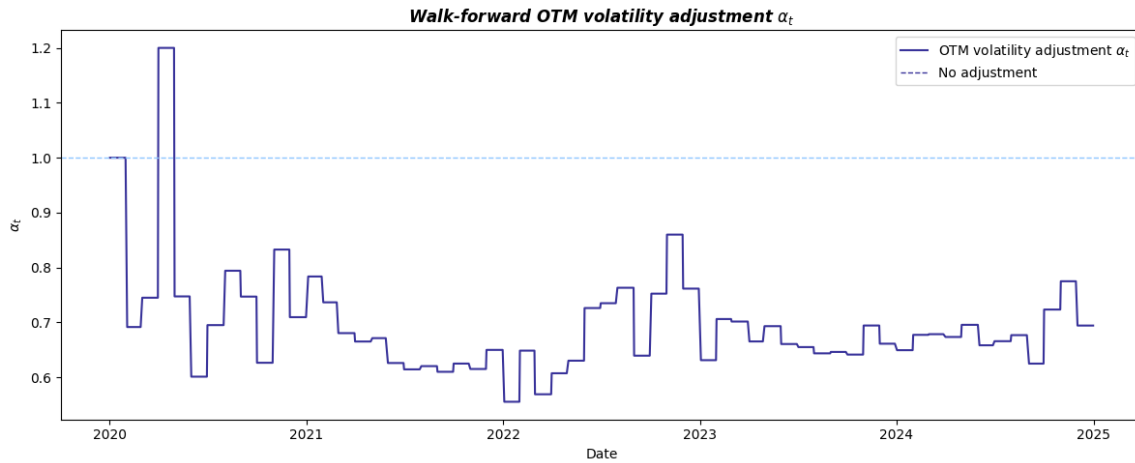


Figure 5.5: Walk-forward OTM volatility adjustment coefficient α_t .

The alpha-adjusted method captures the most significant drawback of the historical model. The price of an OTM option is significantly impacted by the implied option information, skewness, and compensation for crash risk. While a historical HMM can identify volatility regimes, it will not be able to replicate the risk-neutral level of volatility in the price of OTM options without an additional modification to the model.

5.3 Overall interpretation of the OTM exercise

In sum, the OTM valuation results highlight the key weakness of a strictly historical regime-switching approach. Even though RSBS appears to have lower error variance compared to the RV-based strategy, it overvalues OTM call options consistently relative to the squared-error benchmark. In other words, historical volatility regimes cannot be relied on in OTM valuations.

Based on the implied volatility estimates and alpha adjustment, it becomes apparent that risk-neutral levels of volatility, volatility skew, and crash-risk premiums need to be incorporated into the OTM valuation process. Consequently, the RSBS approach should be considered a historical benchmark only.

Chapter 6

Conclusion

The current research examines whether adding regime switching to the Black-Scholes formula allows for improvements in the valuation of financial options with time-varying volatility. The first part of the analysis entailed estimating the Gaussian Hidden Markov Model based on the log-returns of the S&P 500 index. The two-regime model was selected as the operational specification as the two-regime setting allows for a clear and economically interpretable distinction between low- and high-volatility periods while simultaneously avoiding additional estimation instability that would arise from a more nuanced three-regime specification.

Based on the results, it can be concluded that there is a persistence in the behavior of market volatility, low volatility periods exhibit higher persistence than the high volatility ones that appear less often and are usually associated with increased stress in the market. The results support the use of a regime-switching approach, since a single historical-volatility estimate cannot adequately capture time variation in equity-index returns.

In case of at-the-money pricing, the regime-switching Black–Scholes pricing method performs better than the rolling-volatility-based Black-Scholes. Based on the statistical evidence, it can be said that regime switching produces superior results in at-the-money valuations. While the regime switching model does not outperform the benchmark based on every criterion used, it still demonstrates an improvement in pricing.

For the out-of-the money pricing, the performance of the original regime switching model is weaker. Namely, the results show systematic overpricing of OTM calls, especially those priced very low. This result is informative because it shows that the overpricing of OTM call options is revealing the main deficiency of the regime switching model, the lack of calibration on option-implied measures. OTM options are highly sensitive to volatility skewness, tail-risk premium, and liquidity effect among other parameters, which are missing from the HMM estimated using historical prices and realizations of return rates.

The implied-volatility comparison highlights the same limitation. For OTM options, the relevant object is not only the historical volatility estimated from returns, but the Black–Scholes-equivalent implied volatility embedded in market option prices. The walk-forward adjustment confirms that incorporating lagged option-implied information can improve pricing performance.

In summary, however, the findings show that regime switching is beneficial when considering the valuation of options because of its ability to capture variations in market conditions through time. However, regime switching by itself is not sufficient for reproducing an entire option surface. Instead, a fully specified market-consistent model requires both the inclusion of regime switching and some other factor such as risk neutral calibration or volatility skew. In other words, RSBS should be viewed as a historically grounded benchmark rather than as a fully calibrated option-surface model

References

- Boyle, P., and Draviam, T. (2007). *Pricing exotic options under regime switching*. Insurance: Mathematics and Economics, 40(2), 267–282.
- Liew, C. C., and Siu, T. K. (2010). *A hidden Markov regime-switching model for option valuation*. Insurance: Mathematics and Economics, 47(3), 374–384.
- Chan, L., and Zhu, S.-P. (2014). *An exact and explicit formula for pricing lookback options with regime switching*. arXiv working paper.
- Goswami, A., Mukherjee, K. N., Patalwala, I. H., and Sanjay, N. S. (2022). *Regime recovery using implied volatility in Markov modulated market model*. arXiv working paper.
- Rémillard, B., and Rubenthaler, S. (2023). *Option pricing and hedging for regime-switching geometric Brownian motion models*. arXiv working paper.
- Tian, P., Zhou, H., and Zhou, D. (2023). *Analysis about the Black–Scholes asset price under the regime-switching framework*. International Review of Financial Analysis, 88, 102693.
- Pande, N. K., Pasricha, P., Kumar, A., and Gupta, A. K. (2025). *European option pricing in regime switching framework via physics-informed residual learning*. Expert Systems With Applications, 288, 128226.